Instructions for Students: XAI component

Option 1: RenkuLab (cloud)

We will be using **RenkuLab** so everyone has the same environment, without worrying about local setup.

Before you begin you need a **RenkuLab account**. You can sign up with **your email, GitHub acocont** or your institutional **EDU ID**. You do not need a GitHub account for this course.

1. Create your own copy of the course project

1. Go to the course template: https://renkulab.io/p/brandonlpanos/cas-xai-course
2. You will see a banner at the top: **This is a template project**.
   Click **Copy this template**.
3. Give your project a name
4. Your copy now contains:
   → All course notebooks
   → The small dataset
   → The correct Python environment (from requirements.txt)

2. Launch a coding session

1. In your project, scroll to **Sessions** and click ➕.
2. Choose **Create from code** (Renku will detect the requirements.txt).
3. Select **JupyterLab** as the interface.
4. Pick the **default resource class** (enough for this course). If issues then max.
5. Click **Start session**.
   The first build may take a few minutes while dependencies install.
   Budget 15 minutes.

4. Dormant sessions and restarting

Renku automatically pauses sessions that are idle for too long. If this happens, you can restart. Go to your project → Sessions tab. You will see your paused session. Click **Resume** to continue where you left off. If your session becomes terminated after several days, you can always restart **your copy** (go to my coppy in the green banner) of the template.

3. Work on the notebooks

- When your session opens, you will see the notebooks/ folder.
- Open the first notebook and follow the instructions.
- Your changes are saved inside your own Renku project they will not affect the template or others.

5. Saving your work

- Your work is automatically saved in your Renku project.
- If you want to download notebooks, go to **File → Download** in JupyterLab

Option 2: pip virtule enviroment (local)

macOS (Intel & Apple Silicon)
Make a file for your project
$ mkdir <file_name>
Step into the project file
$ cd <file_name>
Clone the repo
$ git clone https://github.com/brandonlpanos/xai-course.git
Move inside the course
$ cd xai-course
Create a virtual environment named cas_env
$ python3 -m venv cas_env
Activate the environment
$ source cas_env/bin/activate
Upgrade pip
$ pip install --upgrade pip
Install the course dependencies from requirements.txt
$ pip install -r requirements.txt
Install Jupyter Notebook
$ pip install notebook ipykernel
Register the kernel so it shows up in Jupyter
python -m ipykernel install --user --name cas_env --display-name "Python (cas_env)"
Launch Jupyter Notebook
$ jupyter notebook
Deactivate later
$ deactivate

Windows (PowerShell recommended)
Clone the repo
$ git clone https://github.com/brandonlpanos/xai-course.git
Move inside the course
$ cd xai-course
Create a virtual environment named cas_env
$ python -m venv cas_env
Activate the environment
$.\cas_env\Scripts\Activate.ps1
If you get an execution policy error, run:
$ Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
$.\cas_env\Scripts\Activate.ps1
Upgrade pip
$ pip install --upgrade pip
Install the course dependencies from requirements.txt
$ pip install -r requirements.txt
Install Jupyter Notebook
$ pip install notebook ipykernel
Register the kernel so it shows up in Jupyter
$ python -m ipykernel install --user --name cas_env --display-name "Python (cas_env)"
Launch Jupyter Notebook
$ jupyter notebook
Deactivate later
$ deactivate

In case of existing repo from last lecture
We do not have to clone again. Simply:
$ cd path/to/xai-course
$ git pull origin main
If your branch is named something else, e.g. master, replace main with that.
$ source cas_env/bin/activate
$ pip install --upgrade pip
$ pip install -r requirements.txt
$ jupyter notebook